

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S8 (R,S) Exam April 2025 (2019 Scheme)

Course Code: CXT418**Course Name: DESIGNING HUMAN CENTERED SYSTEMS****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

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| 1 | What is a human centred system? List out the different methods for observing human experience. | (3) |
| 2 | What are the key principles of participatory research? | (3) |
| 3 | Define WAAD. | (3) |
| 4 | What is the role of people and systems in contextual analysis? | (3) |
| 5 | List the use of design informing models. | (3) |
| 6 | What are Interaction Design Requirements? | (3) |
| 7 | Differentiate design and development. | (3) |
| 8 | What is ideation in design? | (3) |
| 9 | Name three popular modern software tools used for prototyping. | (3) |
| 10 | What is a prototype in design? Why is prototyping important in interaction design? | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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|----|--|-----|
| 11 | a) Explain the concept of contextual inquiry and its role in observing human experience. | (7) |
| | b) How does fly-on-the-wall observation enhance our understanding of human experiences in various contexts? Explain with an example. | (7) |

OR

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| 12 | a) How does walk-a-mile immersion enhance our understanding of human experiences in various contexts? Explain with an example. | (7) |
| | b) Explain the concept of evaluative research and its role in observing human experience. | (7) |

Module II

- 13 a) Explain different methods that will help in developing your ability to support all people and systems. (8)
- b) Explain importance/ difficulty matrix. (6)

OR

- 14 a) Explain the different methods used in finding patterns and determining prioritization. (8)
- b) Explain data interpretation in contextual analysis. (6)

Module III

- 15 a) Explain the general “How to” suggestions for constructing design informing models. (7)
- b) Explain the structure of the requirement statement in the requirement extraction phase with example. (7)

OR

- 16 a) Explain how we could bridge the gap between Analysis and Design. (7)
- b) Explain work environment models with examples. (7)

Module IV

- 17 a) Explain the different methods in design rationale. (7)
- b) Explain the different design perspectives. (7)

OR

- 18 a) What is sketching? Explain essential concepts in sketching. (7)
- b) Explain the different design paradigms. (7)

Module V

- 19 a) Explain the different levels of fidelity in prototypes with examples. (7)
- b) How do prototypes transition from the design phase to becoming a final product, and what challenges may arise during this transition? (7)

OR

- 20 a) Explain the different types of interactive prototypes with examples. (9)
- b) Explain the importance of between-frame transitions. (5)
